

Tab 1

```
#include<stdio.h>  
#include <stdlib.h>
```







































/function to find maximum element

```
int getMax(int arr[],int n){
```

```
int max = arr[0];
```

```
for (int i=1; i<n; i++)  
{
```

```
if(arr[i]>max)
```

```
max = arr[i];  
}  
return max;}
```

```
//Counting Sort based on digit
```

```
void CountingSort (int arr[], int n, int place) {
```

```
int *output = (int *)malloc(n*sizeof(int));
```

```
int count[10]={0};
```

```
//count occurrences
```

```
for (int i = 0; i<n;i++)  
{  
    int digit=(arr[i]/place)%10;  
  
    count[digit]++;  
}
```

```
// cummulative Count
```

```
for (int i=1;i<10;i++)  
{  
count[i]+= count[i-1];  
}
```

```
//Build output array (stable)
```

```
for (int i=n-1; i>=0 ;i--) {
```

```
int digit = (arr[i]/place)%10;
```

```

output[count[digit]-1] = arr[i];

count[digit]--;
}

//copy back to original array

for (int i=0; i<n;i++){
arr[i]=output[i];
}

free(output); // free memory
}

//Radix Sort

void radixSort(int arr[] , int n) {

int Max= getMax(arr,n);

for (int place =1; Max/place >0; place*=10){

CountingSort(arr,n,place);

}

}

//print array

void printArray(int arr[], int n){

for (int i=0; i<n;i++) {

printf("%d ", arr[i]);

}

printf("\n");
}

// Main function

int main(){
int n;

```

```
printf("Enter number of elements:");

scanf("%d",&n);

// Dynamic allocation (no restriction)

int *arr= (int *)malloc(n* sizeof(int));

if(arr == NULL){
printf("Memory allocation failed\n");
return 1;
}

printf(" Enter %d elements:\n", n);

for (int i=0; i<n;i++) {
scanf("%d", &arr[i]);
}

printf("original array : \n");

printArray(arr, n);

radixSort(arr, n);

printf("Sorted array:\n");

printArray(arr, n);

free(arr);

return 0;
}
```

```
Enter number of elements:6
Enter 6 elements:
114 1
115 2
116 3
117 65
118 8
119 9
120 original array :
121 1 2 3 65 8 9
122 Sorted array:
123 1 2 3 8 9 65
124
125 -----
126 (program exited with code: 0)
127 Press return to continue
128 
129 
130
```



Tab 2

